Hall Ticket Number:

MR-22

Code: B425CS5

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. IV Semester Regular/Supplementary Examinations, July 2025

Software Engineering

(Common to CSE, CSB and CSD)

Time: 3 Hours | *Answer ALL Questions* | Max. Marks: 60

Part-A (10 x 1 = 10 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	Define Software Engineering.	1	1	1	1
1. b)	What is spiral Model.	1	2	1	1
1. c)	What is meant by requirement management?	1	1	2	2
1. d)	Define procedural interfaces.	1	1	2	2
1. e)	List the goals of the design process.	1	2	3	3
1. f)	What is UML? Write the principles of modelling.	1	1	3	3
1. g)	What is integration testing?	1	1	4	4
1. h)	Define black box testing strategy.	1	1	4	4
1. i)	Define software reliability.	1	1	5	4
1. j)	What is software risk?	1	1	5	4

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	Construct the CMMI model with a neat sketch	5	3	1	1
2. b)	Discuss on various types of software myths and the true aspects of these myths.	5	4	1	1
	OR				
2. c)	Compare and contrast different software process models.	5	4	1	1
2. d)	Explain process frame work activities.	5	2	1	1
3. a)	Explain the difference between functional requirements and non-functional requirements	5	4	2	2
3. b)	Demonstrate in detail about system requirements.	5	2	2	2
	OR				
3. c)	State and explain various aspects in requirements validation process.	5	2	2	2
3. d)	Explain in detail about requirements elicitation and analysis process.	5	2	2	2
4. a)	Identify the major elements in pattern based software design in detail.	5	3	3	3
4. b)	Explain collaboration diagrams and use case diagrams.	5	2	3	3
	OR				
4. c)	Draw and explain Use case diagrams and its development process with suitable illustrations	5	3	3	3
4. d)	Discuss about architectural design in brief.	5	2	3	3
5. a)	Explain in detail about white box testing techniques.	5	2	4	4
5. b)	Discuss about metrics for design model and source code.	5	3	4	4
	OR				
5. c)	Describe strategic approach to software testing.	5	2	4	4
5. d)	Explain about metrics for design model.	5	2	4	4
6. a)	Write a detailed note on ISO 9000 quality standards.	5	2	5	4
6. b)	What is statistical software quality Assurance? Discuss its objectives.	5	3	5	4
	OR				
6. c)	Explain about risk projection and risk management.	5	3	5	4
6. d)	List and explain the various software quality factors.	5	2	5	4

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome